Marco Biroli, Research Scholar

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Google Scholar

Education

2025 – Now **Research Scholar, University of Chicago.**
Department of Physics, Eric and Wendy Schmidt Scholarship for AI in Science, Chicago.
Supervisor: Vincenzo Vitelli.
Research focus: *multidisciplinary approaches to statistical physics, biological systems and machine learning.*

2022 – 2025 **Ph.D., Université Paris-Saclay.**
Laboratoire de Physique Théorique et de Modèles Statistiques (LPTMS), Orsay.
Supervisor: Satya N. Majumdar.
Thesis title: *Interacting particles with stochastic resetting*

2020 – 2022 **M.Sc. ICFP in theoretical physics** ENS-PSL, Paris.
GPA: 17.96/20

2019 – 2020 **Licence 3 in Physics** ENS-PSL, Paris.
GPA: 15.5/20 in Physics and 19.5/20 in Mathematics and Computer Science.

2017 – 2020 **Bachelor of Science, double major in physics and mathematics** Ecole Polytechnique.
Summa Cum Laude. GPA: 4.17/4.0

Awards

On hold. **Kadanoff-Rice Postdoctoral Scholarship.**
Competitive 1 Year PostDoctoral Fellowship.

2025 – 2027 **Eric and Wendy Schmidt Scholarship for AI in Science**, 169 744 $
Competitive 2 Year PostDoctoral Fellowship.

2022 – 2025 **Contrats doctoraux spécifiques normaliens (CDSN)**, 75 600 €
Competitive 3 Year Doctoral Scholarship.

2019 – 2022 **Scholarship of the physics departement of ENS-PSL**, ∼ 10 000 €.
Competitive 3 year excellence scholarship.

2018 – 2020 **Excellence Scholarship from the Ecole Polytechnique Foundation**, 9 600 €.
Competitive 2 year excellence scholarship.

Teaching

January 2026 – Now **Supervision** University of Chicago.
Supervision of a graduate student of Vincenzo Vitelli.

Teaching (continued)

September 2024 – June 2025 **Mentoring** Université Paris-Saclay.
Supervision of a first year Ph.D. student of Satya N. Majumdar.

March 2024 – June 2024 **Assistant Professor** Université Paris-Saclay.
Undergraduate Experimental Physics

6-16 October 2023 **National divulgation physics event aimed at highschool students**
Animator of a stand on Probabilities.

2019 – 2022 **Private Tutoring**

Conferences, Workshops & Seminars

3 March 2026 **Variational Auto-Encoders are finite-size mean-field approximators**
Schmidt AI in Science Speaker Series
Seminar Speaker

17-19 June 2024 **New Vistas in Stochastic Resetting**
Higgs Centre for Theoretical Physics
Contributed Poster

March 2024 **Talk at the LPTMS bi-weekly seminar series**
Invited Talk.

3-7 April 2023 **Workshop "Interaction, Disorder and Elasticity"**
Ecole de Physique des Houches
Contributed Talk

28-30 November 2022 **Groupement de Recherche (GDR) "Interaction, Disorder and Elasticity"**
Université Grenobles-Alpes
Contributed Poster

3-5 October 2022 **Asymmetry in interacting particle systems, microscopic and macroscopic effects**
INRIA Lille

Schools

July 2024 **Theory of Large Deviations and Applications summer school**
Ecole de physique des Houches.
Contributed Poster

July 2023 **Quantum localization and Glassy physics**
Institut d'études scientifiques de Cargèse
Contributed Poster

June 2023 **Statistical Physics of Complex Systems**
The Beg Rohu Summer School
Contributed Poster

Languages & Computer skills

- English (Proficient), French (Mother tongue), Italian (Mother tongue), Spanish (B2)
- Python, Pytorch, C++, HTML, CSS, Javascript

Hobbies

- Climbing, mountaineering, skiing, paragliding, sailing, fencing.

Miscellaneous

2022-Now **Reviewer**
Review for Physical Review Letters,Physical Review E, J-Phys A, J-Stat, Chaos, Chaos Solitons & Fractals.

2024-Now **LPTMS Student Representative at the lab board**
Elected to be the student representative at the General Assembly of the laboratory.

2022 – 2024 **Assessor for the Bachelor of Ecole Polytechnique admissions**
Assessed around 50 candidate files for admission.

2022-2023 **Creator and organizer of the LPTMS Statistical Physics seminar series**
Created a seminar series and took care of inviting the speakers as well as organizing the logistic of the seminar.